Case Study 2: Progressive Electricity Billing

1. Problem statement

Develop a Python program to calculate a household's electricity bill based on the number of electricity units consumed. The program should apply different rates for different consumption slabs, add a fixed charge, calculate the final bill, and display the billing details.

2. Objectives

- To accept electricity consumption from the user.
- To calculate charges using progressive electricity slabs.
- To add fixed charges to the bill.
- To display the total electricity bill.
- To practice variables, arithmetic operators, conditional statements, and user input in Python.

3. Assumptions

Since your case study gives no specific electricity tariff rates, we'll use these sample rates for this assignment:

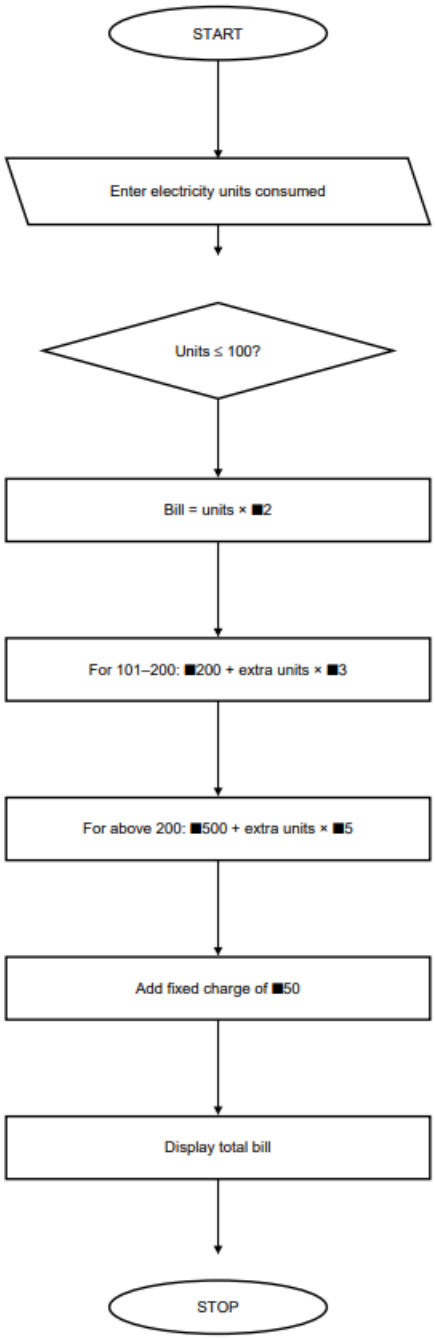
Electricity consumption	**Rate per unit**
First 100 units	₹2
Next 100 units (101–200)	₹3
Above 200 units	₹5

- Fixed charge: ₹50 per bill.
- The user enters a non-negative number of units.
- Each slab is charged at its own rate, rather than applying one rate to all units.
- No additional taxes, rebates, or surcharges are included in this basic version.

4. Algorithm

1. Start the program.
2. Accept the number of electricity units consumed from the user.
3. Check whether the units entered are negative.
4. If the input is invalid, display an error message.
5. If consumption is up to 100 units, calculate the charge at ₹2 per unit.
6. If consumption is between 101 and 200 units, charge ₹2 for the first 100 units and ₹3 for the remaining units.
7. If consumption exceeds 200 units, charge ₹2 for the first 100 units, ₹3 for the next 100 units, and ₹5 for the remaining units.
8. Add the fixed charge of ₹50.
9. Display the unit consumption, energy charge, fixed charge, and total bill.
10. Stop the program.

5. Flowchart



6. Pseudocode

START

INPUT units

IF units < 0 THEN DISPLAY "Invalid electricity units" ELSE IF units <= 100 THEN energy_charge = units * 2

ELSE IF units <= 200 THEN energy_charge =

(100 * 2) + ((units - 100) * 3)

ELSE energy_charge = (100 * 2) +

(100 * 3) + ((units - 200) * 5)

END IF

fixed_charge = 50 total_bill =

energy_charge + fixed_charge

DISPLAY units DISPLAY energy_charge DISPLAY fixed_charge DISPLAY total_bill END IF

STOP

7. Source code

electricity bill.py

```
units = int(input("Enter electricity units consumed: "))

if units < 0:
    print("Invalid input! Units cannot be negative.")

else:
    if units <= 100:
        energy_charge = units * 2

    elif units <= 200:
        energy_charge = (100 * 2) + ((units - 100) * 3)

    else:
        energy_charge = (100 * 2) + (100 * 3) + ((units - 200) * 5)

    fixed_charge = 50
    total_bill = energy_charge + fixed_charge

    print("\nELECTRICITY BILL")
    print("Units Consumed:", units)
    print("Energy Charge: Rs.", energy_charge)
    print("Fixed Charge: Rs.", fixed_charge)
    print("Total Bill: Rs.", total_bill)
```

8. Test cases

Input(units)	Expected output
0	Total bill= ₹50
100	Total bill= ₹250
-10	Invalid input

9. Output

```
"C:\Users\tejas\PycharmProjects\Python Activity\.venv\
Enter electricity units consumed: 0

ELECTRICITY BILL
Units Consumed: 0
Energy Charge: Rs. 0
Fixed Charge: Rs. 50
Total Bill: Rs. 50

Process finished with exit code 0
```

```
"C:\Users\tejas\PycharmProjects\Python Activity\.venv\Scri
Enter electricity units consumed: -10
Invalid input! Units cannot be negative.

Process finished with exit code 0
```

```
"C:\Users\tejas\PycharmProjects\Python Activity\.ve
Enter electricity units consumed: 100

ELECTRICITY BILL
Units Consumed: 100
Energy Charge: Rs. 200
Fixed Charge: Rs. 50
Total Bill: Rs. 250

Process finished with exit code 0
```

10. Conclusion

The Progressive Electricity Billing program calculates a household's electricity bill based on the number of units consumed. It applies different rates to different consumption slabs, adds a fixed charge, and displays the final bill. This program demonstrates the use of user input, variables, arithmetic operators, and conditional statements in Python.